Tyler Barna

EDUCATION

Ph.D. Astrophysics **Expected 2027**
Data Science in Astrophysics Minor
University of Minnesota-Twin Cities

M.S. Astrophysics **2024**
Data Science in Astrophysics Minor
University of Minnesota-Twin Cities

B.S. Astrophysics **May 2021**
Rutgers University, New Brunswick - Honors College

RESEARCH

Graduate Research Assistant **2023-Present**
Goddard Space Flight Center
Project: *GCN: General Coordinates Network*

- Contributing to development of the search functionality for the GCN Circulars Archive, including the addition of the use of natural language processing to improve search quality and extract information from the historic archive
- Work with GCN will continue through at least 2026

Graduate Research Assistant **2021-Present**
Department of Physics and Astronomy, University of Minnesota-Twin Cities
Advisor: Assistant Professor Michael Coughlin
Project: *NMMA: A Nuclear-Physics and Multi-Messenger Astrophysics Framework*

- Working with large collaboration to study kilonovae via gravitational wave detection and electromagnetic observation
- Leading development of pipeline to automate analysis of ZTF data to inform follow-up observations of potential gravitational wave companions

Senior Honors Thesis **2020-2021**
Department of Physics and Astronomy, Rutgers University
Advisor: Professor Saurabh Jha
Project: *Early-Time Observations of Type Ia Supernovae with TESS*

- Conducted independent research on SNe Ia observed by the Transiting Exoplanet Survey Satellite at early rise times - Project received "High Honors"
- Engaged in monthly remote observation shifts on the Keck II Telescope

Undergraduate Researcher **2018-2020**
Department of Astrophysical Sciences, Princeton University
Advisor: Dr. Melinda Soares-Furtado

- Worked both locally and remotely on several research projects, including hydrodynamic simulations of globular clusters and analysis of data from the Kepler Space Telescope's K2 Mission
- Used time-management skills and conducted independent research on novel topics to independently accomplish project goals in a timely manner

Governor's School Scholar **2016**
New Jersey Governor's School of the Sciences, Drew University
Advisor: Dr. David Cincotta
Project: *Controlled Release Kinetics*

- Collaborated with a large group of students from across New Jersey to complete a complex research project in chemistry on a contracted time scale
- Helped to automate the collection of data and designed an experimental procedure that greatly increased the number of trials that could be run within the given time constraints

SELECTED PUBLICATIONS

ZTF25abjmnps (AT2025ulz) and S250818k: A candidate superkilonova from a subthreshold subsolar gravitational-wave trigger
Kasliwal, M.M.; Ahumada, T.; Stein, R.; ... **Barna, T.**; et al. 2025, ApJL, Volume 995, Issue 2. (arXiv:2510.23732)

IIb or not IIb: A Catalog of ZTF Kilonova Imposters
Barna, T.; Fremling, C.; Ahumada, T.; et al. 2025, PASP 137(8), 084105. (arXiv:2506.15900)

LIGO/Virgo/KAGRA neutron star merger candidate S250206dm: Zwicky Transient Facility observations
Ahumada, T.; Anand, S.; Bulla, M.; ... **Barna, T.**; et al. 2025, arXiv eprint (in review). (2507.00357)

Searching for Gravitational Wave Optical Counterparts with the Zwicky Transient Facility: Summary of O4a
Ahumada, T.; Anand, S.; Coughlin, M.W.; ... **Barna, T.**; et al. PASP 136(11), 114201. (2405.12403)

An online framework for fitting fast transient light curves
Barna, T.; Reed, B.; Coughlin, M.W., et al. 2024, MNRAS 531(1), 1084-1094. (arxiv:2404.17515

An updated nuclear-physics and multi-messenger astrophysics framework for binary neutron star mergers
Pang, P.T.H.; Dietrich, T.; Coughlin, M.W.; ... **Barna, T.**; et al. 2023, Nature Commun. 14 (2023) 1, 8352. (arXiv:2205.08513)

SALT3-NIR: Taking the Open-Source Type Ia Supernova Model to Longer Wavelengths for Next-Generation Cosmological Measurements
Pierel, J.D.R.; Jones, D.O.; Kenworthy, W.D.; ... **Barna, T.**; et al. 2022, arXiv eprint. (arXiv:2209.05594)

SN 2021csp – the explosion of a stripped envelope star within a H and He-poor circumstellar medium
Fraser, M.; Stritzinger, M.D.; Brennan, S.J.; ... **Barna, T.**; et al. 2021, arXiv eprint. (arXiv:2108.07278)

A Catalog of Periodic Variables in Open Clusters M35 and NGC 2158
Soares-Furtado, M.; Hartman, J.D.; Bhatti, W.; Bouma, L.G.; **Barna, T.**; Bakos, G.A. 2020, ApJS, Volume 246, Issue 1. (arXiv:1911.00832)

CONFERENCES

ID4 AI for Engineering Workshop **2026**

247th Meeting of the American Astronomical Society **2026**

Recent and Planned Development of The General Coordinates Network (Poster)

AstroRob VIII: Workshop on Autonomous Robotic Observatories **2025**
The General Coordinates Network: NASA's Next Generation Time-Domain and Multi-Messenger Alert System (Talk)

245th Meeting of the American Astronomical Society **2025**
IIb or not IIb: Identifying Kilonova Imposters (poster)

Hotwiring the Transient Universe VII **2024**
Optimization of Kilonova Follow-up with a Multi-Armed Bandit Strategy (Talk)

Numanji/NMMA Collaboration Retreat **2024**

243rd Meeting of the American Astronomical Society **2024**
Development of the General Coordinates Network Circulars Search Functionality (Poster)

ZTF Summer School **2022, 2023, 2024, 2025**
Coordination Assistant for NMMA (Talk)

NASA Kepler and K2 SciCon V **2019**
Searching for Exoplanets in Dense Clusters with Image Subtraction (Poster)

HONORS & AWARDS

Robert O. Pepin Fellow	**Summer 2026**
Visiting Scholar at Italian National Institute for Astrophysics	**Spring 2026**
Data Science in Multi-Messenger Astrophysics Research Fellowship	**Spring 2025**
Winner of UMN Departmental 3 Minute Thesis Competition	**2022, 2023**
Data Science in Multi-Messenger Astrophysics Research Fellowship	**2022-2023**
Ney Fellowship	**2022**
Rutgers University Trustee Scholarship	**2017-2021**
Hoffman DiMuzio Community Scholarship	**2017**
Brett Fenster Memorial Scholarship	**2017**

COMPUTER SKILLS

Languages and Frameworks: Python, Java, JavaScript, LaTeX, React, Remix
Applications: Anaconda, Mathematica, Git, Windows Subsystem for Linux, Slurm, WinSCP, PuTTY, Adobe Creative Suite, Microsoft Office
Operating Systems: Unix, Linux, Windows, Android

PROFESSIONAL AFFILIATIONS

Accelerated AI Algorithms for Data-Driven Discovery (A3D3) **2022-Present**
Graduate Student Trainee

American Astronomical Society **2018-Present**
Student Member

WORK EXPERIENCE

USSF Certified Soccer Referee **2015-2021**
South Jersey Soccer League & South Jersey Girls Soccer League

- Utilized thorough knowledge of the laws of the game to quickly and decisively make calls with consistency and in a manner appropriate for the level of play
- Worked with other referees to make quick decisions in games ranging from adolescent to collegiate levels

Red Cross Certified Lifeguard **2016-2019**
Del Val Pool Maintenance
- Measured and recorded chemical levels on an hourly basis, making adjustments to satisfy safety standards
- Maintained pool site and supervised swimmers

Personal tutor **2015-2017**
Self-Employed
- Provided tutoring to students between K-8 in various topics, particularly STEM

OUTREACH

MIfA Public Observing Nights	**2022-Present**
Astronomy on Tap - Twin Cities	**2022-2023**
Coughlin Group Elementary School Outreach	**2022-2023**
ZTF Summer School Program	**2022-Present**
MIfA Universe@Home Program	**2021-2022**
Rutgers Astronomical Society	**2018-2021**
Rutgers Honors College Mentoring Program	**2017-2018**

Interests

INTERESTS & HOBBIES

Computer Building, Cooking, Guitar, History, Literature, Lutherie, Music